

Week – 4 Assignment

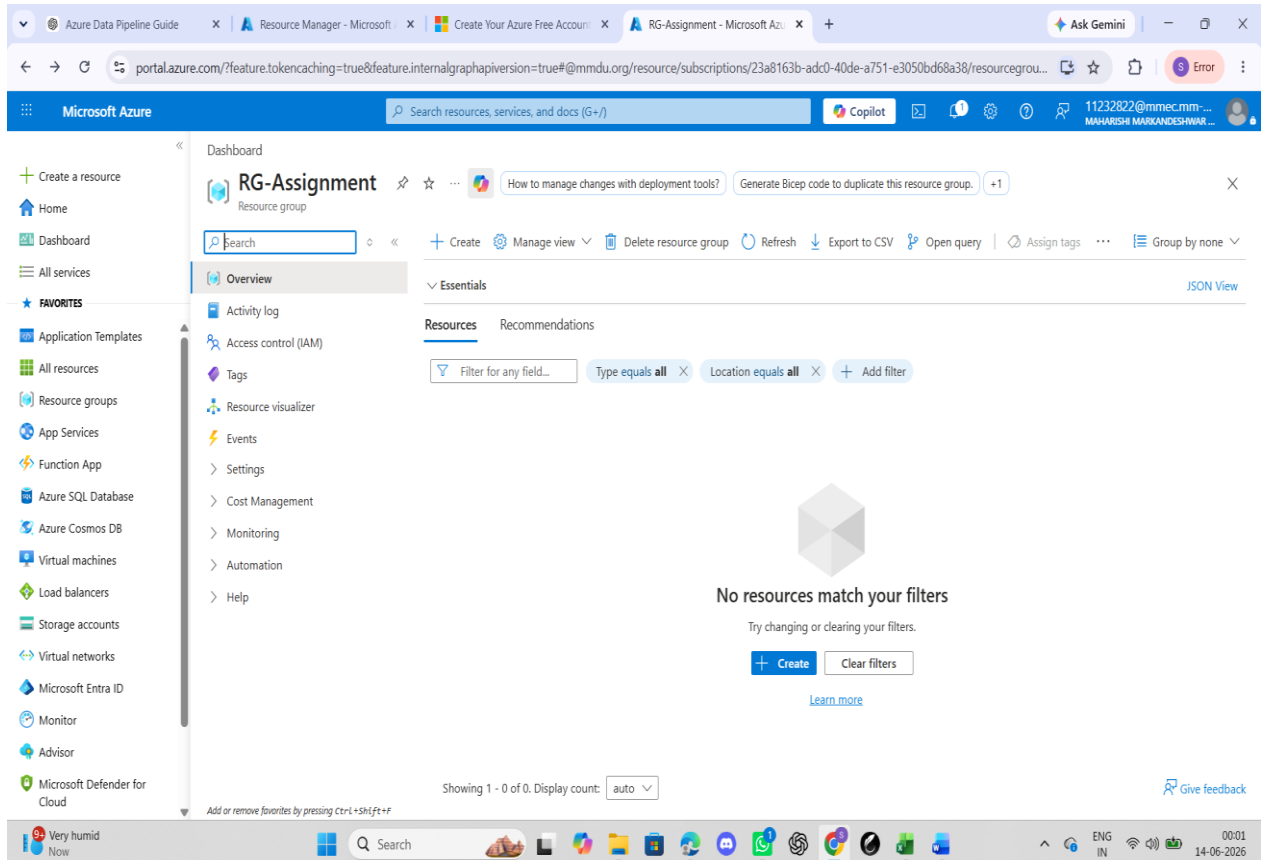
Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

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1. Objective

Understand Azure cloud concepts and build an end-to-end data pipeline using Storage Account and Azure Data Factory. Steps: Explore Azure Portal and create a Resource Group. Create a Storage Account, Blob Container, and upload a CSV file. Create Azure Data Factory and explore ADF UI (Author, Monitor, Manage). Create Linked Service (Blob Storage) and datasets (source and destination). Use Get Metadata activity to retrieve file information. Build a pipeline using Copy Data activity and configure source and destination. Execute the pipeline using Debug/Trigger and monitor execution. Assign IAM roles (Reader, Contributor) and ensure access between ADF and Storage. Implement an end-to-end pipeline (Blob → ADF → Destination) with metadata validation.

2. Resource Group Creation



3. Storage Account Creation

Dashboard > Storage center | Blob Storage

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags Review + create

[View automation template](#)

Basics

Subscription	Azure subscription 1
Resource group	RG-Assignment
Location	Central India
Storage account name	sameepstorage822
Primary service	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Managed identities for CMR	Disabled

[Previous](#) [Next](#) [Create](#)

[Give feedback](#)

4. Blob Container Creation

Dashboard > Storage center | Blob Storage > sameepstorage822

sameepstorage822 | Containers

Storage account

[Add container](#) [Upload](#) [Refresh](#) [Delete](#) [Change access level](#) [Reset](#)

Showing all 1 items

☐	Name	Last modified	Anonymous access level
☐	\$logs	6/14/2026, 12:05:25 AM	Private

New container

Name *

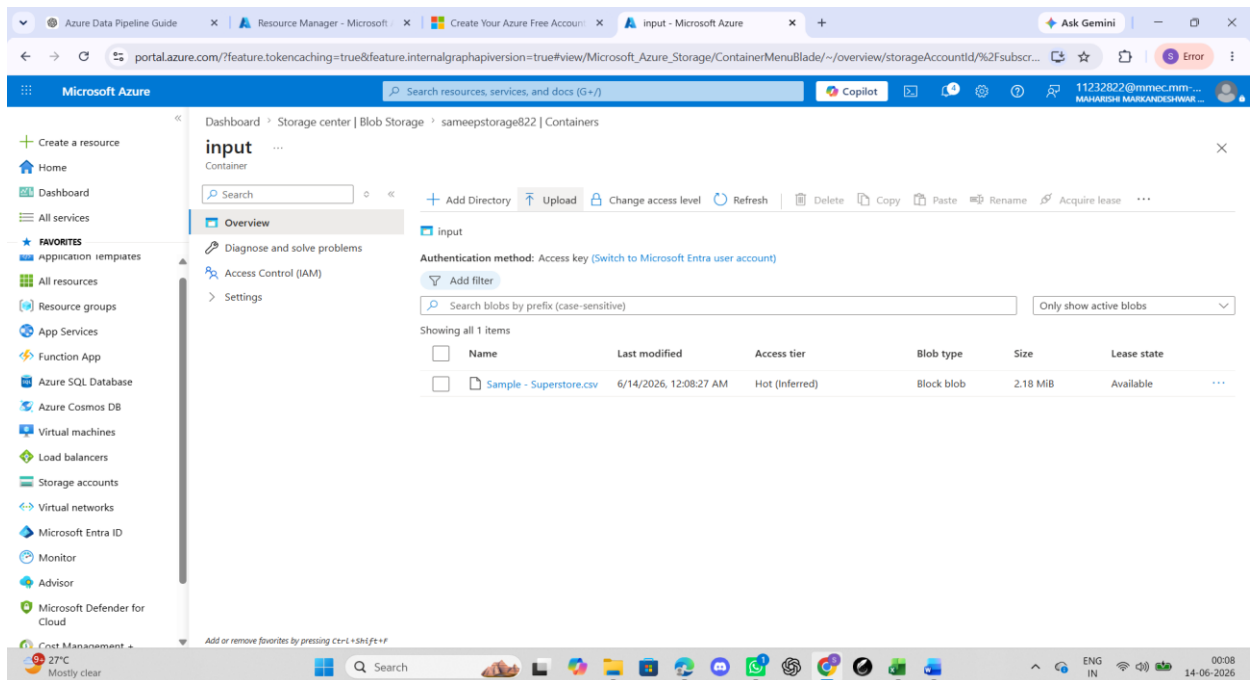
Anonymous access level

[The access level is set to private because anonymous access is disabled on this storage account.](#)

[Advanced](#)

[Create](#) [Give feedback](#)

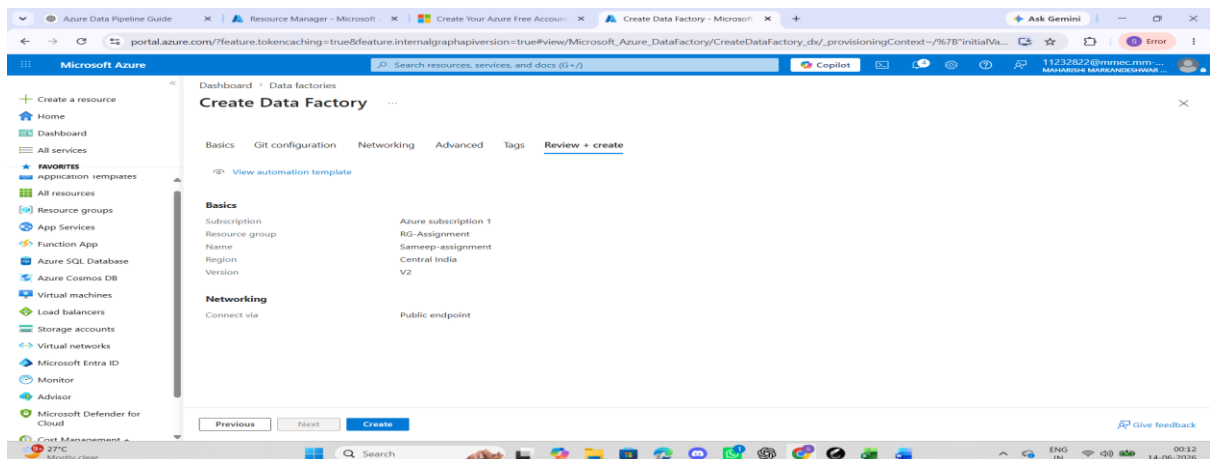
5. CSV File Upload



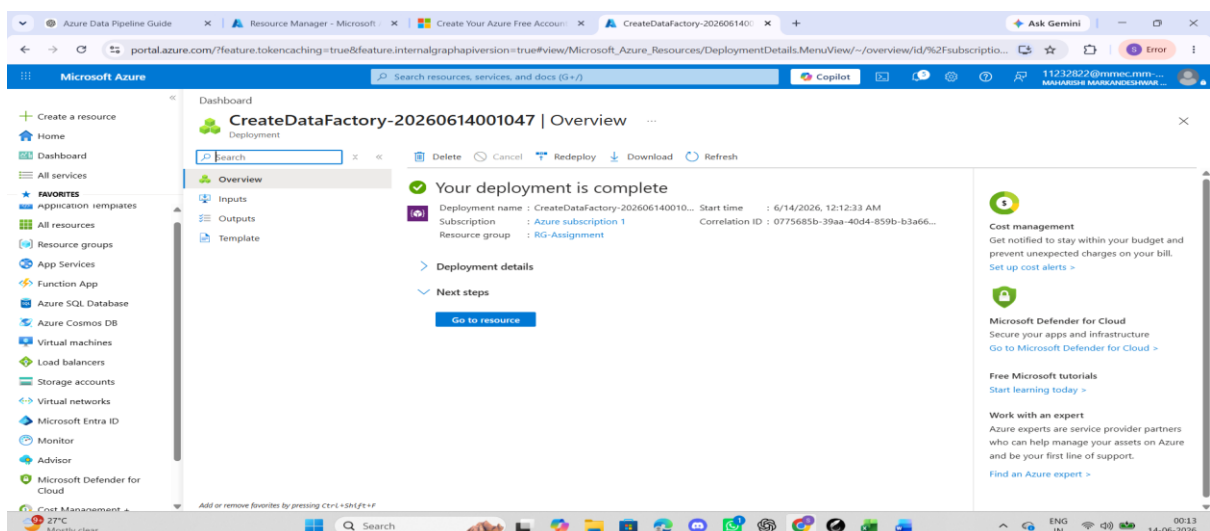
The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation links for 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the 'input' container overview. The container is named 'input' and is located under 'sameepstorage822'. It contains one item, 'Sample - Superstore.csv', which is a 2.18 MiB block blob uploaded on 6/14/2026 at 12:08:27 AM. The authentication method is 'Access key'.

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/14/2026, 12:08:27 AM	Hot (Inferred)	Block blob	2.18 MiB	Available

6. Azure Data Factory Creation

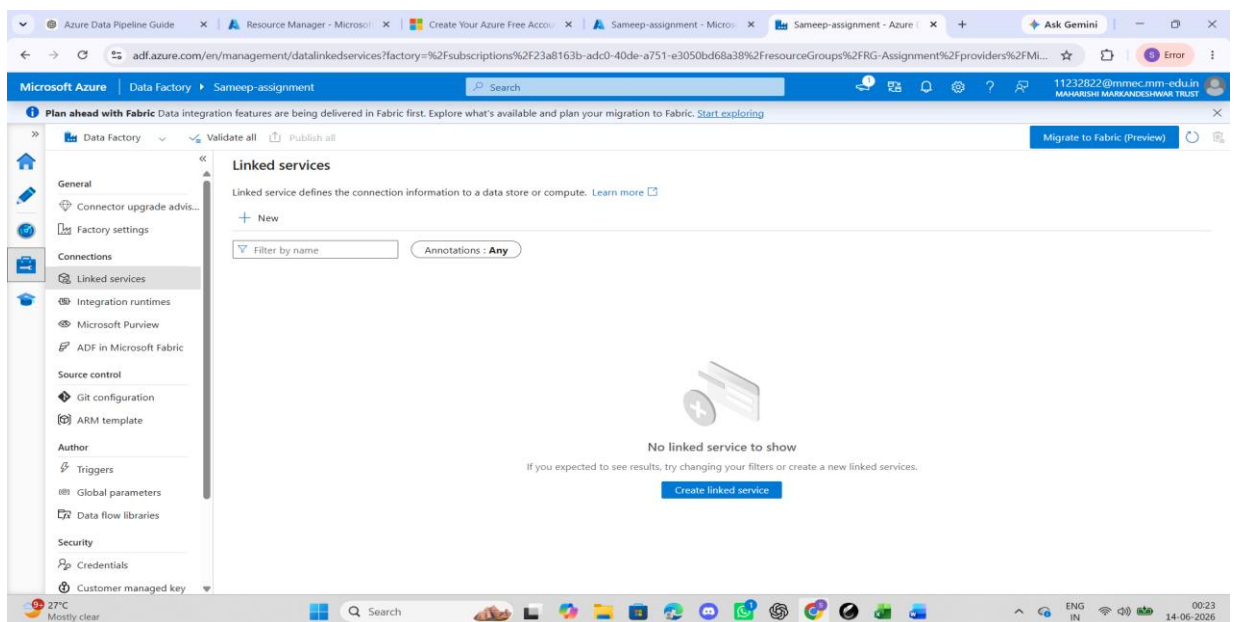
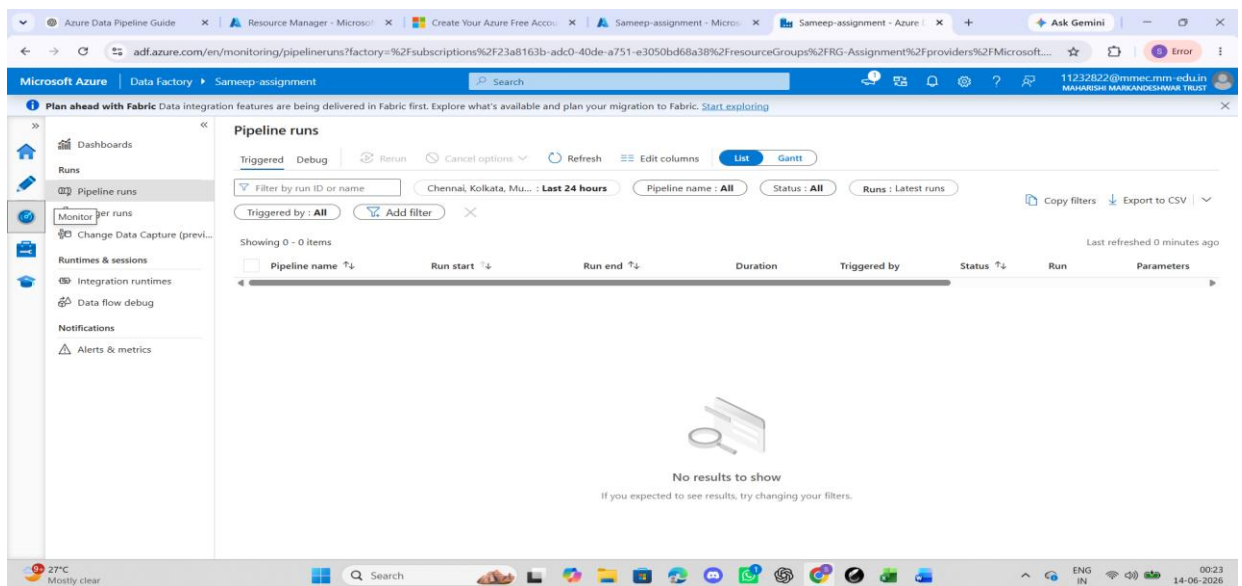
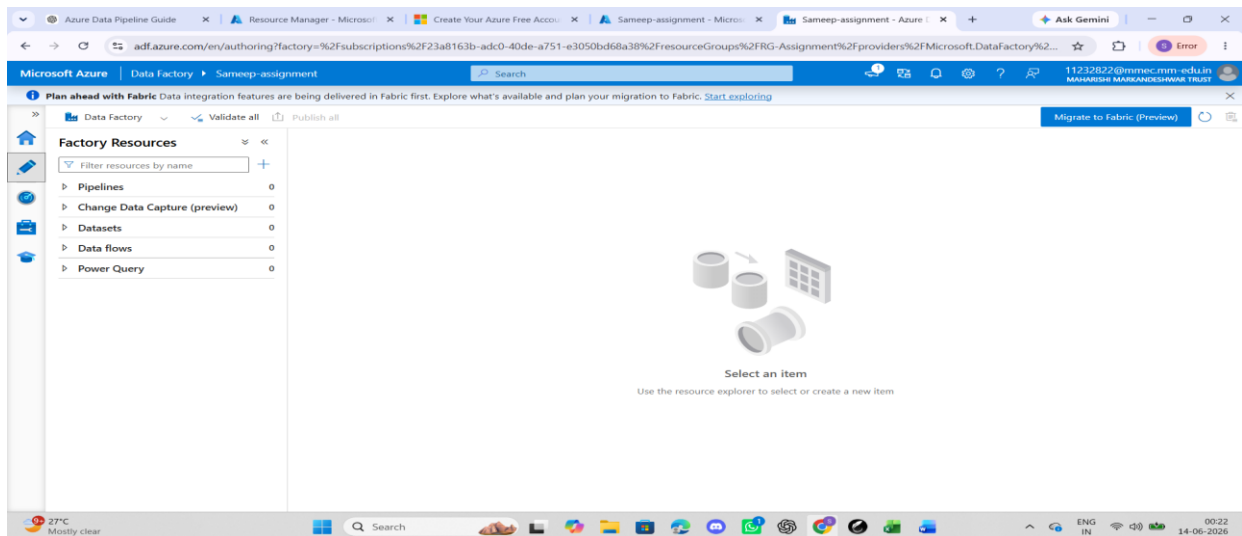


The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation links for 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the 'Create Data Factory' wizard. The 'Basics' tab is selected, showing the subscription 'Azure subscription 1', resource group 'Sameep-assignment', region 'Central India', and version 'V2'. The 'Networking' tab shows 'Connect via' as 'Public endpoint'.



The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation links for 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area displays the 'CreateDataFactory-20260614001047' deployment overview. The deployment is complete, showing the deployment name, subscription, and resource group. The 'Next steps' section includes 'Go to resource'.

7. ADF Studio Overview



8. Linked Service Configuration

The screenshot shows the 'New linked service' configuration page in the Microsoft Azure Data Factory portal. The left sidebar contains navigation options like 'General', 'Connections', 'Integration runtimes', 'Source control', 'Author', 'Security', and 'Customer managed key'. The main area is titled 'Linked services' and shows a 'New' button and a 'Filter by name' input. The right pane is titled 'New linked service' and shows the configuration for 'Azure Blob Storage'. The 'Name' field is 'LS_BlobStorage'. The 'Description' field is empty. The 'Connect via integration runtime' dropdown is set to 'AutoResolveIntegrationRuntime'. The 'Authentication type' dropdown is set to 'Account key'. The 'Connection string' tab is selected, and the 'Account selection method' is set to 'From Azure subscription'. The 'Azure subscription' dropdown is set to 'Select all'. The 'Storage account name' dropdown is set to 'sameepstorage822'. The 'Additional connection properties' section has a '+ New' button. At the bottom, there are 'Create', 'Back', 'Test connection', and 'Cancel' buttons.

9. Source Dataset Creation

The screenshot shows the 'DelimitedText1' dataset configuration page in the Microsoft Azure Data Factory portal. The left sidebar contains navigation options like 'Factory Resources', 'Pipelines', 'Change Data Capture (preview)', 'Datasets', 'Data flows', and 'Power Query'. The main area is titled 'DelimitedText1' and shows a 'CSV' icon. The 'Properties' pane on the right shows the 'Name' field as 'DelimitedText1' and the 'Description' field as empty. The 'Connection' tab is selected, and the 'Linked service' dropdown is set to 'LS_BlobStorage'. The 'File path' field is set to 'input / Directory / Sample - Superstore.csv'. The 'Compression type' dropdown is set to 'No compression'. The 'Column delimiter' dropdown is set to 'Comma (,)'.

10. Pipeline Creation

The screenshot displays the Microsoft Azure Data Factory portal. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'PL_CopyData'. The main canvas shows the 'PL_CopyData' pipeline with a single activity named 'DelimitedText1'. The 'Activities' pane on the right lists various activity types, including 'Move and transform', 'Synapse', 'Azure Data Explorer', 'Azure Function', 'Batch Service', 'Databricks', 'General', 'HDInsight', 'Iteration & conditionals', 'Machine Learning', and 'Power Query'. The 'Properties' pane on the right shows the 'General' tab for the selected activity, with fields for 'Name' (PL_CopyData) and 'Description'. The top navigation bar includes 'Data Factory' and 'Sameep-assignment'.

11. Get Metadata Activity

The screenshot displays the Microsoft Azure Data Factory portal, focusing on the 'Get Metadata' activity configuration. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'PL_CopyData'. The main canvas shows the 'PL_CopyData' pipeline with a single activity named 'Get Metadata1'. The 'Activities' pane on the right lists various activity types, including 'General', 'Get Metadata', 'Move and transform', 'Synapse', 'Azure Data Explorer', 'Azure Function', 'Batch Service', 'Databricks', 'General', 'HDInsight', 'Iteration & conditionals', 'Machine Learning', and 'Power Query'. The 'Properties' pane on the right shows the 'General' tab for the selected activity, with fields for 'Name' (PL_CopyData) and 'Description'. The 'Settings' tab is active, showing the 'Dataset' dropdown set to 'DelimitedText1' and the 'Field list' section with 'Argument', 'Exists', and 'Size' selected. The top navigation bar includes 'Data Factory' and 'Sameep-assignment'.

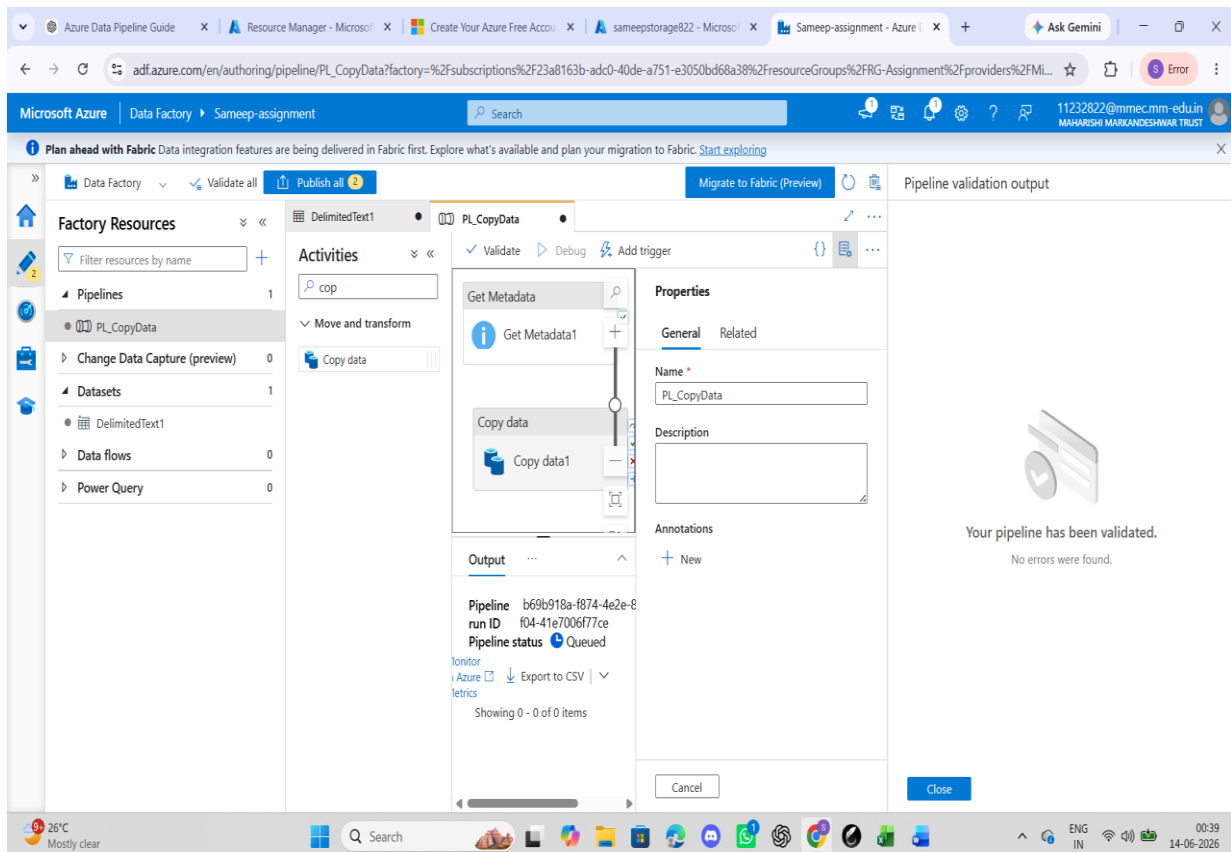
12. Copy Data Activity

The screenshot displays the Microsoft Azure Data Factory console interface. The top navigation bar shows the user is logged in as 11232822@mmeccmm.edu.in. The main content area is titled 'Data Factory > Sameep-assignment'. On the left, the 'Factory Resources' pane shows a list of resources including Pipelines (1), Datasets (1), Data flows (0), and Power Query (0). The 'PL_CopyData' pipeline is selected. The 'Activities' pane shows a search for 'cop' and a list of activities including 'Get Metadata' and 'Copy data'. The 'Copy data' activity is selected, and its configuration is shown in the 'Settings' tab. The 'Concurrency' is set to 1, and the 'Elapsed time metric' is set to 0.00:10:00. The 'Properties' pane on the right shows the 'General' tab with the 'Name' set to 'PL_CopyData' and a 'Description' field. A 'Cancel' button is visible at the bottom right of the settings pane.

13. Validate

The screenshot displays the Microsoft Azure Data Factory console interface, showing the validation of a pipeline. The top navigation bar shows the user is logged in as 11232822@mmeccmm.edu.in. The main content area is titled 'Data Factory > Sameep-assignment'. On the left, the 'Factory Resources' pane shows a list of resources including Pipelines (1), Datasets (1), Data flows (0), and Power Query (0). The 'PL_CopyData' pipeline is selected. The 'Activities' pane shows a search for 'cop' and a list of activities including 'Get Metadata' and 'Copy data'. The 'Copy data' activity is selected, and its configuration is shown in the 'Settings' tab. The 'Concurrency' is set to 1, and the 'Elapsed time metric' is set to 0.00:10:00. The 'Properties' pane on the right shows the 'General' tab with the 'Name' set to 'PL_CopyData' and a 'Description' field. A 'Cancel' button is visible at the bottom right of the settings pane. On the right side of the console, a 'Pipeline validation output' pane shows a message: 'Your pipeline has been validated. No errors were found.' with a 'Close' button.

14. Debug



Microsoft Azure | Data Factory | Sameep-assignment

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines: 1
 - PL_CopyData
- Change Data Capture (preview): 0
- Datasets: 1
 - DelimitedText1
- Data flows: 0
- Power Query: 0

Activities

- cop
- Move and transform
 - Copy data

Get Metadata

- Get Metadata1

Copy data

- Copy data1

Properties

General

Name *

PL_CopyData

Description

Annotations

Output

Pipeline: b69b918a-f874-4e2e-8f04-41e7006f77ce

run ID: f04-41e7006f77ce

Pipeline status: Queued

Monitor | Azure | Export to CSV | Metrics

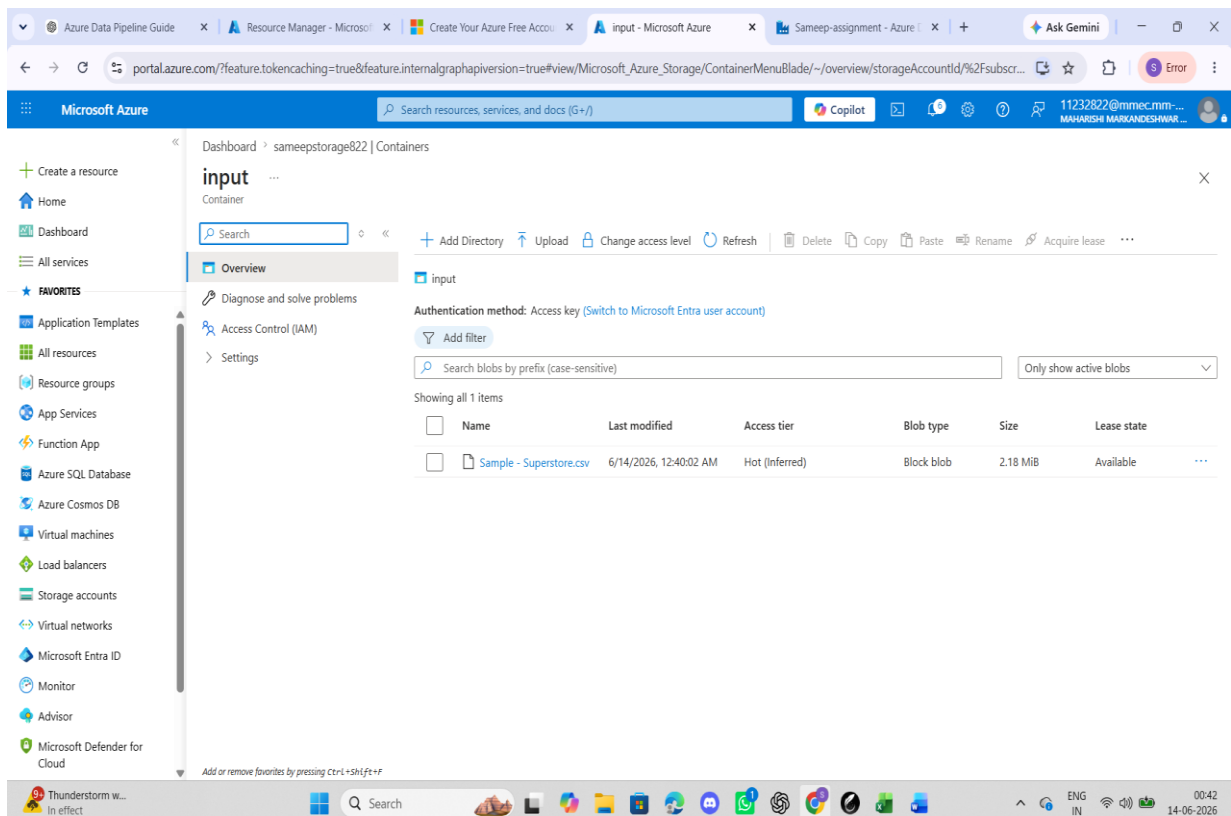
Showing 0 - 0 of 0 items

Pipeline validation output

Your pipeline has been validated.

No errors were found.

15. Output Check



Microsoft Azure | Dashboard | sameepstorage822 | Containers

input

Container

Search

+ Add Directory | Upload | Change access level | Refresh | Delete | Copy | Paste | Rename | Acquire lease

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

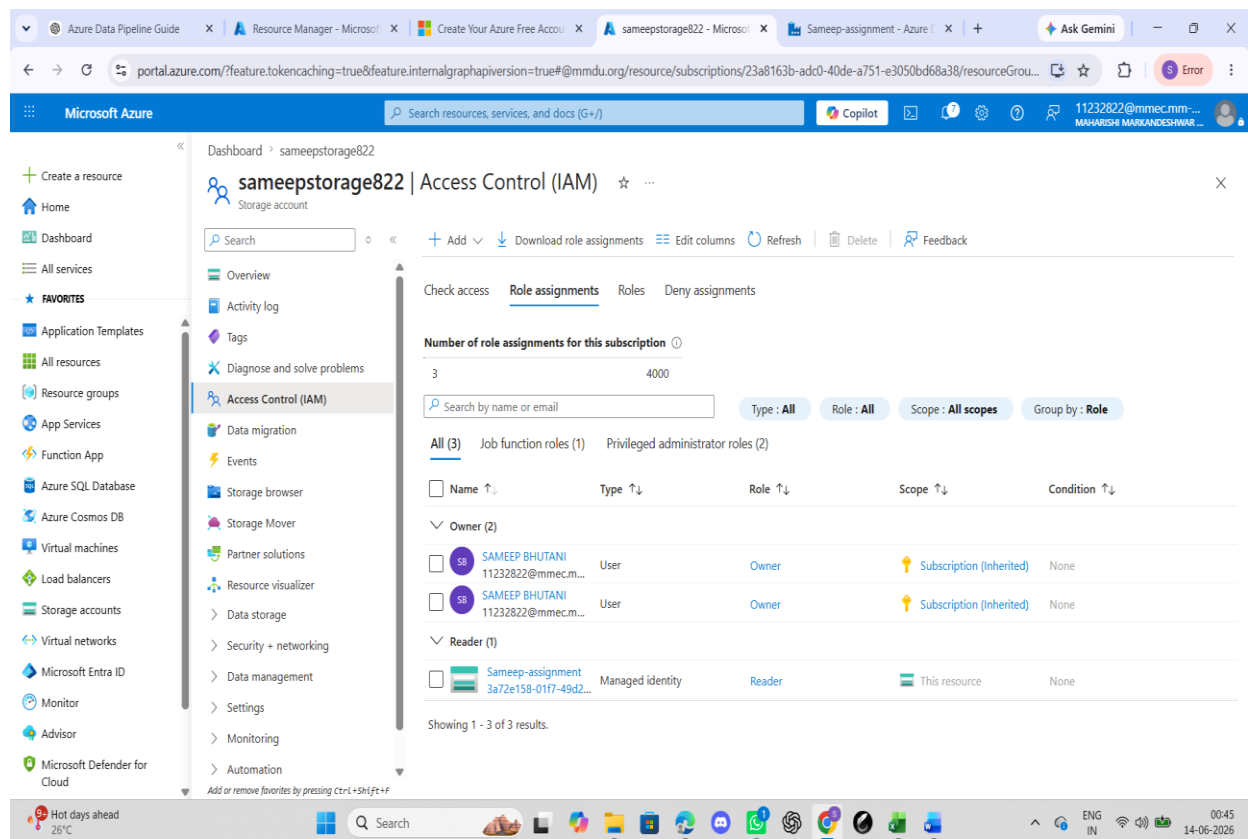
Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/14/2026, 12:40:02 AM	Hot (inferred)	Block blob	2.18 MiB	Available

16. IAM Role Assignment



17. Result

The pipeline executed successfully and copied the CSV file from the source container to the destination container.

18. Conclusion

This assignment successfully demonstrated the implementation of an end-to-end data pipeline using Microsoft Azure services. A Resource Group, Storage Account, Blob Containers, and Azure Data Factory were created and configured. The provided CSV file was uploaded to Azure Blob Storage and connected to Azure Data Factory through Linked Services and Datasets. The Get Metadata activity was used to validate and retrieve information about the source file, while the Copy Data activity was used to transfer the file from the source container to the destination container. The pipeline was executed successfully using the Debug and Trigger options, and the execution status was monitored through the Monitor section of Azure Data Factory. Proper access permissions were also ensured through IAM role assignments. The successful creation and execution of the pipeline confirmed the seamless integration between Azure Storage and Azure Data Factory for data movement and processing.